

RBE/CS 549: Computer Vision

Project 2 – Buildings built in minutes - SfM and NeRF - Phase 1 Report

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Abstract—We present a complete classical incremental Structure from Motion (SfM) pipeline which was applied to a five-image outdoor sequence of Unity Hall at Worcester Polytechnic Institute. Starting from pre-computed SIFT correspondences, every stage of the pipeline is implemented from scratch in Python: RANSAC-based outlier rejection across all image pairs, fundamental and essential matrix estimation via the normalised 8-point algorithm, four-hypothesis pose extraction, linear and non-linear triangulation, incremental camera registration via PnP-RANSAC and non-linear PnP refinement, and sparse bundle adjustment. The pipeline successfully registered all five cameras and recovered a sparse 3-D model of the building structure, along with the respective camera poses. Several non-trivial implementation issues were identified and resolved, including correct pixel-space RANSAC thresholding, cheirality-based DLT sign disambiguation, and gauge-freedom removal in bundle adjustment. The final reconstruction is qualitatively consistent with the scene geometry and is supported by visualization figures produced at each pipeline stage.

I. INTRODUCTION

Structure from Motion (SfM) simultaneously recovers a sparse three-dimensional scene model and the camera poses from a set of two-dimensional images. It is a foundational technique in computer vision with applications in autonomous driving, augmented reality and robotics.

This report presents a complete implementation of a classical incremental SfM pipeline applied to five images of Unity Hall at WPI. Pre-computed SIFT keypoints and descriptors are provided; every subsequent stage is implemented from scratch in Python following the algorithmic template in Algorithm 1.

The pipeline consists of nine stages:

- 1) Feature matching and outlier rejection via RANSAC (all pairs)
- 2) Fundamental matrix estimation (normalised 8-point algorithm)
- 3) Essential matrix computation and correction
- 4) Camera pose extraction (four hypotheses from $\mathbf{E}$)
- 5) Linear triangulation for the bootstrap image pair
- 6) Cheirality-based pose disambiguation
- 7) Non-linear triangulation refinement (Levenberg–Marquardt)

- 8) Incremental camera registration via PnP-RANSAC + Non-linear PnP
- 9) Sparse bundle adjustment (gauge-fixed, with reprojection gating)

Algorithm 1 Incremental SfM pipeline (Algorithm 3)

```

all image pairs  $(i, j)$   $[\mathbf{x}_i, \mathbf{x}_j]$   $\leftarrow$ 
GETINLIERSRANSAC( $\mathbf{x}_i, \mathbf{x}_j$ )  $\mathbf{F}$   $\leftarrow$ 
ESTIMATEFUNDAMENTALMATRIX( $\mathbf{x}_1, \mathbf{x}_2$ )
 $\mathbf{E} \leftarrow$  ESSENTIALMATRIX( $\mathbf{F}, \mathbf{K}$ )  $\{(\mathbf{C}_k, \mathbf{R}_k)\}_{k=1}^4 \leftarrow$ 
EXTRACTCAMERAPOSE( $\mathbf{E}$ )  $k = 1 : 4$   $\mathbf{X}_k \leftarrow$ 
LINEARTRIANGULATION( $\mathbf{K}, \mathbf{0}, \mathbf{I}, \mathbf{C}_k, \mathbf{R}_k, \mathbf{x}_1, \mathbf{x}_2$ )
 $(\mathbf{C}_2, \mathbf{R}_2) \leftarrow$  DISAMBIGUATECAMERAPOSE( $\{(\mathbf{C}_k, \mathbf{R}_k, \mathbf{X}_k)\}$ )
 $\mathbf{X} \leftarrow$  NONLINEARTRIANGULATION( $\mathbf{K}, \dots, \mathbf{X}_0$ )
 $i = 3 : I$   $(\hat{\mathbf{C}}, \hat{\mathbf{R}}) \leftarrow$  PNPRANSAC( $\mathbf{X}, \mathbf{x}_i, \mathbf{K}$ )
 $(\mathbf{C}_i, \mathbf{R}_i) \leftarrow$  NONLINEARPnP( $\mathbf{X}, \mathbf{x}_i, \mathbf{K}, \hat{\mathbf{C}}, \hat{\mathbf{R}}$ )
 $\mathbf{X}_{\text{new}} \leftarrow$  LINEARTRIANGULATION( $\dots$ )
 $\mathbf{X}_{\text{new}} \leftarrow$  NONLINEARTRIANGULATION( $\dots, \mathbf{X}_{\text{new}}$ )
 $\mathbf{X} \cup= \mathbf{X}_{\text{new}}$   $\mathbf{V} \leftarrow$  BUILDVISIBILITYMATRIX(tracks)
 $(\{\mathbf{C}_k\}, \{\mathbf{R}_k\}, \mathbf{X}) \leftarrow$  BUNDLEADJUSTMENT( $\dots, \mathbf{K}, \mathbf{V}$ )

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II. DATASET AND CAMERA MODEL

A. Dataset

Five images (800×600 px) of Unity Hall at WPI were captured with a Samsung S22 Ultra (f/1.8, ISO 50, 1/500 s) and pre-rectified to remove lens distortion. Pre-computed SIFT matches are stored in `matchingi.txt` files, where the matching is between camera i and $i+1$.

B. Pinhole Camera Model

A 3-D world point $\mathbf{X} \in \mathbb{R}^3$ projects to pixel $\mathbf{x} \in \mathbb{R}^2$ via

$$\lambda \tilde{\mathbf{x}} = \mathbf{P} \tilde{\mathbf{X}}, \quad \mathbf{P} = \mathbf{K} \mathbf{R} [\mathbf{I} \mid -\mathbf{C}] \quad (1)$$

where $\tilde{\cdot}$ denotes homogeneous coordinates, $\mathbf{K}$ is the 3×3 intrinsic matrix, $\mathbf{R} \in SO(3)$ the camera rotation, $\mathbf{C} \in \mathbb{R}^3$ the camera center in world coordinates, and $\lambda > 0$ the projective depth. The translation vector relates as $\mathbf{t} = -\mathbf{R}\mathbf{C}$.

The calibration matrix for this dataset is:

$$\mathbf{K} = \begin{pmatrix} 531.12 & 0 & 407.19 \\ 0 & 531.54 & 313.31 \\ 0 & 0 & 1 \end{pmatrix} \quad (2)$$

Optical axis in world coordinates. The camera's optical axis points along $+Z_{\text{cam}}$; converting to world coordinates yields:

$$\mathbf{fwd} = \mathbf{R}^\top \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \mathbf{r}_3 \quad (3)$$

where $\mathbf{r}_3$ is the *third* row of $\mathbf{R}$, which is used to disambiguate camera poses using the positive depth constraint.

III. FEATURE MATCHING AND RANSAC

A. Feature Tracks

Matching files are parsed to build a global track dictionary $\{\text{track_id} \mapsto \{(\text{img_id}, u, v)\}\}$. Tracks link the same physical point observed across multiple views and are essential for propagating 3-D point indices during incremental registration. Coordinate lookup uses rounding to 1 decimal place (0.1 px tolerance) to avoid missed matches from floating-point formatting differences in the files.

B. Fundamental Matrix – Normalised 8-Point Algorithm

The *fundamental matrix* $\mathbf{F}$ (3×3 , rank 2) encodes the epipolar constraint:

$$\tilde{\mathbf{x}}_2^\top \mathbf{F} \tilde{\mathbf{x}}_1 = 0 \quad (4)$$

For m correspondences this gives the linear system $\mathbf{A}\mathbf{f} = \mathbf{0}$, where each row of $\mathbf{A} \in \mathbb{R}^{m \times 9}$ is:

$$[x_1x'_1, x_1y'_1, x_1, y_1x'_1, y_1y'_1, y_1, x'_1, y'_1, 1] \quad (5)$$

The solution $\mathbf{f} = \text{vec}(\mathbf{F})$ is the right singular vector of $\mathbf{A}$ corresponding to the smallest singular value. Rank-2 is enforced by zeroing the smallest singular value of $\mathbf{F}$: $\mathbf{F} = \mathbf{U}_F \text{diag}(\sigma_1, \sigma_2, 0) \mathbf{V}_F^\top$.

a) Isotropic normalisation.: Points are normalised before solving by the transform

$$\mathbf{T} = \begin{pmatrix} s & 0 & -s\bar{u} \\ 0 & s & -s\bar{v} \\ 0 & 0 & 1 \end{pmatrix}, \quad s = \frac{\sqrt{2}}{\sigma_{\text{pts}}} \quad (6)$$

and denormalised afterwards: $\mathbf{F} = \mathbf{T}_2^\top \hat{\mathbf{F}} \mathbf{T}_1$.

C. RANSAC Outlier Rejection

Algorithm 2 GETINLIERSRANSAC

```

 $n \leftarrow 0, S_{\text{in}} \leftarrow \emptyset$ 
for  $i = 1 : M$ 
  Sample 8 correspondences randomly
   $\mathbf{F} \leftarrow \text{ESTIMATEFUNDAMENTALMATRIX}(\cdot)$ 
   $j = 1 : N$ 
  if  $d_j < \varepsilon$ 
     $S \leftarrow S \cup \{j\}$ 
  if  $|S| > n$ 
     $n \leftarrow |S|$ 
     $S_{\text{in}} \leftarrow S$ 

```

The inlier criterion uses the *epipolar constraint*:

$$d_i = |\tilde{\mathbf{x}}_{2i}^\top \mathbf{F} \tilde{\mathbf{x}}_{1i}| \quad (7)$$

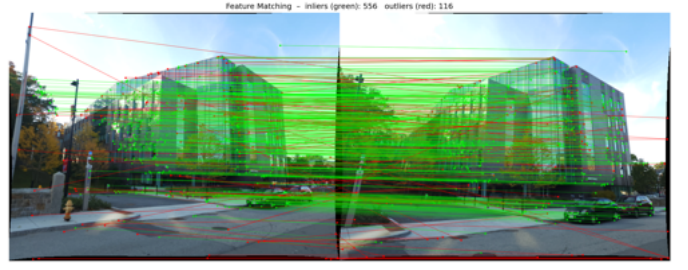


Fig. 1. Feature matching for the bootstrap pair (images 1 & 2). Green lines: 556 RANSAC inliers. Red lines: 116 outliers rejected by the symmetric epipolar constraint ($\varepsilon = 0.05$ px).

D. Epipolar Geometry

Fig. 2 illustrates the difference between a rank-3 and a rank-2 fundamental matrix. The top row shows $\mathbf{F}_{\text{rank-3}}$ (constructed by setting $\sigma_3 = 0.5\sigma_2$): lines do not converge because the null space is no longer a single point. The bottom row shows the correctly enforced rank-2 $\mathbf{F}$: all epipolar lines converge at the epipole $\mathbf{e}$, marked with a yellow arrow (image 1: $\mathbf{e} \approx (1155, 440)$; image 2: $\mathbf{e} \approx (1038, 424)$).



Fig. 2. Epipolar lines. **Top:** rank-3 $\mathbf{F}$ (no convergence). **Bottom:** rank-2 $\mathbf{F}$ (lines converge at epipole $\mathbf{e}$, yellow arrow).

IV. ESSENTIAL MATRIX AND CAMERA POSE

A. Essential Matrix

$$\mathbf{E} = \mathbf{K}^\top \mathbf{F} \mathbf{K} \quad (8)$$

The singular values of $\mathbf{E}$ are corrected to $(1, 1, 0)$:

$$\mathbf{E} \leftarrow \mathbf{U} \text{diag}(1, 1, 0) \mathbf{V}^\top \quad (9)$$

Unlike $\mathbf{F}$ (pixel space, 7 DoF), $\mathbf{E}$ operates in normalised image coordinates and has 5 DoF.

B. Extracting Camera Poses

Let $\mathbf{E} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$ and $\mathbf{W} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. The four pose hypotheses are:

$$\mathbf{C}_1 = +\mathbf{u}_3, \quad \mathbf{R}_1 = \mathbf{U}\mathbf{W}\mathbf{V}^\top \quad (10)$$

$$\mathbf{C}_2 = -\mathbf{u}_3, \quad \mathbf{R}_2 = \mathbf{U}\mathbf{W}\mathbf{V}^\top \quad (11)$$

$$\mathbf{C}_3 = +\mathbf{u}_3, \quad \mathbf{R}_3 = \mathbf{U}\mathbf{W}^\top\mathbf{V}^\top \quad (12)$$

$$\mathbf{C}_4 = -\mathbf{u}_3, \quad \mathbf{R}_4 = \mathbf{U}\mathbf{W}^\top\mathbf{V}^\top \quad (13)$$

where $\mathbf{u}_3$ is the third column of $\mathbf{U}$. If $\det(\mathbf{R}) < 0$, correct with $\mathbf{R} \leftarrow -\mathbf{R}$, $\mathbf{C} \leftarrow -\mathbf{C}$.

V. TRIANGULATION

A. Linear Triangulation (DLT)

For two cameras with projection matrices $\mathbf{P}_1, \mathbf{P}_2$, each correspondence gives two linear constraints. The 4×4 system is:

$$\mathbf{A} = \begin{pmatrix} x_1\mathbf{P}_1^{(3)} - \mathbf{P}_1^{(1)} \\ y_1\mathbf{P}_1^{(3)} - \mathbf{P}_1^{(2)} \\ x_2\mathbf{P}_2^{(3)} - \mathbf{P}_2^{(1)} \\ y_2\mathbf{P}_2^{(3)} - \mathbf{P}_2^{(2)} \end{pmatrix}, \quad \mathbf{A}\tilde{\mathbf{X}} = \mathbf{0} \quad (14)$$

The solution $\tilde{\mathbf{X}}$ is the last right singular vector of $\mathbf{A}$, normalised so that $\tilde{X}_4 = 1$.

B. Cheirality Condition and Pose Disambiguation

A point $\mathbf{X}$ is in front of camera $(\mathbf{C}_i, \mathbf{R}_i)$ if:

$$\mathbf{r}_3^{(i)} \cdot (\mathbf{X} - \mathbf{C}_i) > 0 \quad (15)$$

For camera 1 ($\mathbf{C}_1 = \mathbf{0}$, $\mathbf{R}_1 = \mathbf{I}$) this simplifies to $X_z > 0$. Disambiguation selects the hypothesis maximising the number of triangulated points satisfying (15) for *both* cameras simultaneously.

C. Non-Linear Triangulation

The linear estimate is refined by minimising total reprojection error:

$$\min_{\mathbf{X}} \sum_{j=1,2} \left[\left(u^j - \frac{\mathbf{P}_1^{j\top} \tilde{\mathbf{X}}}{\mathbf{P}_3^{j\top} \tilde{\mathbf{X}}} \right)^2 + \left(v^j - \frac{\mathbf{P}_2^{j\top} \tilde{\mathbf{X}}}{\mathbf{P}_3^{j\top} \tilde{\mathbf{X}}} \right)^2 \right] \quad (16)$$

solved with Levenberg–Marquardt (`scipy.optimize.least_squares, method='lm'`).

VI. PERSPECTIVE-N-POINT (PNP)

A. Linear PnP (DLT)

Given $N \geq 6$ 2-D/3-D correspondences, image points are first normalised: $\hat{\mathbf{x}} = \mathbf{K}^{-1}\tilde{\mathbf{x}}$. The $2N \times 12$ design matrix satisfies $\mathbf{A} \text{vec}(\mathbf{P}) = \mathbf{0}$; the solution is the last row of $\mathbf{V}^\top$ from the SVD of $\mathbf{A}$. Decomposing $\mathbf{M} = \mathbf{P}_{:,1:3} = \mathbf{U}_M \mathbf{D}_M \mathbf{V}_M^\top$ gives $\mathbf{R} = \mathbf{U}_M \mathbf{V}_M^\top$ where the determinant is corrected to +1.

The DLT solution is scale-ambiguous: both $+\mathbf{P}$ and $-\mathbf{P}$ satisfy the normal equations. Both signs are tested and the one yielding the majority of 3-D points in front of the camera, using cheirality, is kept. This fix resolved camera-4 pose collapse, where the uncorrected sign placed the camera center at the mirror position behind the scene.

B. PnP-RANSAC

Algorithm 3 PNPRANSAC

```

 $n \leftarrow 0$ ,  $S_{\text{in}} \leftarrow \emptyset$   $i = 1 : M$  Sample 6 correspondences
randomly  $(\hat{\mathbf{C}}, \hat{\mathbf{R}}) \leftarrow \text{LINEARPnP}(\cdot)$   $j = 1 : N$   $e \leftarrow$ 
 $\|\mathbf{x}_j - \pi(\hat{\mathbf{C}}, \hat{\mathbf{R}}, \mathbf{X}_j)\|^2$   $e < \varepsilon_r^2$   $S \leftarrow S \cup \{j\}$   $|S| > n$ 
 $n \leftarrow |S|$ ;  $S_{\text{in}} \leftarrow S$  Re-estimate  $(\hat{\mathbf{C}}, \hat{\mathbf{R}})$  from all inliers in
 $S_{\text{in}}$ 

```

Threshold $\varepsilon_r = 29$ px was used, as smaller values gave zero inliers for cameras 4–5.

C. Non-Linear PnP

Pose is refined by minimising reprojection error using quaternion parameterisation to implicitly enforce $\mathbf{R} \in SO(3)$:

$$\min_{\mathbf{C}, \mathbf{q}} \sum_{i=1}^J \left[\left(u^i - \frac{\mathbf{P}_1^{i\top} \tilde{\mathbf{X}}_i}{\mathbf{P}_3^{i\top} \tilde{\mathbf{X}}_i} \right)^2 + \left(v^i - \frac{\mathbf{P}_2^{i\top} \tilde{\mathbf{X}}_i}{\mathbf{P}_3^{i\top} \tilde{\mathbf{X}}_i} \right)^2 \right] \quad (17)$$

Initial guess $(\mathbf{C}_0, \mathbf{R}_0)$ comes from PnP-RANSAC.

VII. BUNDLE ADJUSTMENT

A. Visibility Matrix

A binary $I \times J$ matrix $\mathbf{V}$ is constructed where $V_{ij} = 1$ if point j is visible from camera i :

$$V_{ij} = \begin{cases} 1 & \text{camera } i \text{ observes point } j \\ 0 & \text{otherwise} \end{cases} \quad (18)$$

B. Sparse Bundle Adjustment

The Bundle Adjustment (BA) objective jointly optimises all camera poses and 3-D point positions:

$$\min_{\{\mathbf{C}_i, \mathbf{q}_i\}, \{\mathbf{X}_j\}} \sum_{i=1}^I \sum_{j=1}^J V_{ij} \left[\left(u^j - \frac{\mathbf{P}_1^{j\top} \tilde{\mathbf{X}}_j}{\mathbf{P}_3^{j\top} \tilde{\mathbf{X}}_j} \right)^2 + \left(v^j - \frac{\mathbf{P}_2^{j\top} \tilde{\mathbf{X}}_j}{\mathbf{P}_3^{j\top} \tilde{\mathbf{X}}_j} \right)^2 \right] \quad (19)$$

a) *Parameterisation*: Each camera has 7 parameters $[q_x, q_y, q_z, q_w, C_x, C_y, C_z]$; each 3-D point contributes 3. With $I = 5$ cameras and J points the parameter vector has dimension $35 + 3J$.

b) *Gauge fixing*: Without constraints, the objective (19) has 7 gauge degrees of freedom (global rigid motion + scale), causing the optimiser to collapse all cameras to a single point. Camera0 is frozen throughout BA: its 7 parameters are constant and contribute residual rows but zero Jacobian columns.

c) *Reprojection gating*: Observations with initial reprojection error > 20 px are excluded before BA to prevent tracks contaminated by early bad poses from distorting the global estimate.

d) *Jacobian sparsity*: Observation (i, j) affects only camera i 's 7 parameters and point j 's 3 parameters. The sparsity pattern is pre-computed from $\mathbf{V}$, making Jacobian evaluation scale as $O(|\mathcal{V}|)$ rather than $O((35 + 3J)^2)$. `scipy.optimize.least_squares (method='trf')` with `ftol=10-4`, `gtol=10-6` is used throughout.

VIII. RESULTS

A. Camera Disambiguation and Triangulation

Of the four pose hypotheses extracted from **E**, two (blue and orange in Fig. 3) place the majority of reconstructed points at negative Z values or at large distances inconsistent with the scene scale, reflecting the mirror ambiguity inherent in the essential matrix decomposition. The remaining two hypotheses (green and red) cluster compactly near the origin with positive Z , and the cheirality test unambiguously selects the one that maximises the number of points in front of both cameras simultaneously.

The selected hypothesis is then refined by non-linear triangulation. As shown in Fig. 4, the linear (red) and non-linear (blue) point clouds are nearly indistinguishable at this scale, which is expected: the linear DLT estimate is already consistent with the epipolar geometry, and the LM step makes only small corrections to individual point depths. Camera 1 is fixed at the origin by construction; camera 2 sits at approximately $(0.9, 0, 0)$, confirming the predominantly lateral baseline between the first two images. The reconstructed points form a characteristic V-shape tracing the two wings of the building, a structure that persists and grows clearer throughout the incremental pipeline.

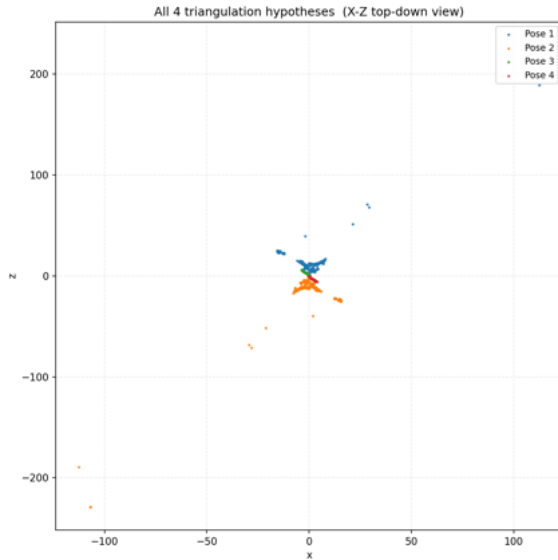


Fig. 3. All 4 triangulation hypotheses (X-Z top-down view). No camera markers are drawn because pose is not yet disambiguated.

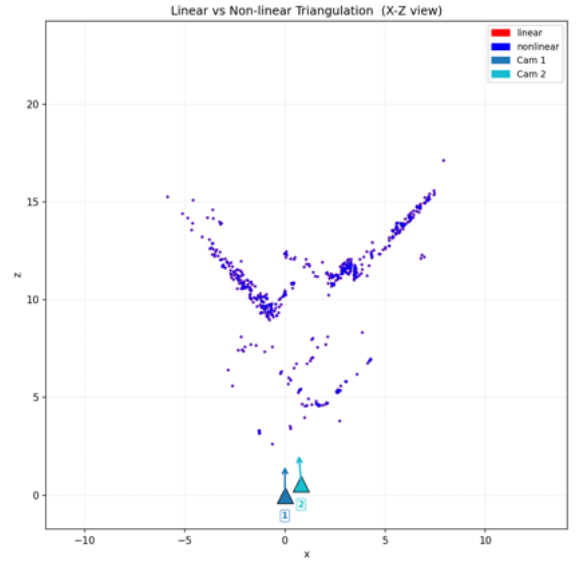


Fig. 4. Linear (red) vs. non-linear (blue) triangulation for the bootstrap pair. The two clouds are nearly identical, confirming the linear estimate provides a good initialisation for the LM refinement.

B. Bootstrap Reprojection

A direct measure of triangulation quality is the reprojection error: the pixel-space distance between each observed feature and the projection of its corresponding 3-D point. For the bootstrap pair, the reprojected crosses (red) fall tightly on top of the observed feature locations (green) across the full extent of the building facade in both images, as shown in Fig. 5. The non-linear result (bottom row) shows marginally tighter clustering relative to the linear result (top row), with the improvement most visible near the corners and edges of the building where perspective distortion is largest. This confirms that the bootstrap reconstruction is geometrically sound and provides a reliable seed for the incremental registration that follows.



Fig. 5. Reprojection overlay for the bootstrap pair (images 1 & 2). Green: observed. Red $\times$: reprojected. Top: linear triangulation. Bottom: non-linear triangulation.

C. Incremental Camera Registration

All three remaining cameras are successfully registered, with the recovered camera centers tracing a smooth arc consistent with a sweep around the building corner (Table I). Camera 3 poses an interesting edge case: only 6 of 329 PnP correspondences survive the RANSAC inlier test, yet the non-linear refinement converges to a geometrically plausible pose at (1.682, 0.228, 1.213). The very low inlier count reflects the relatively small overlap between image 3 and the bootstrap pair, combined with noise in the early 3-D cloud; the key insight is that 6 well-distributed inliers are sufficient to determine the 6-DoF pose, and the subsequent LM step corrects for the imprecision of the RANSAC-only estimate. Cameras 4 and 5 benefit from the denser cloud accumulated after camera 3's triangulation, yielding 535 and 486 inliers respectively, robust enough to produce accurate initial poses before non-linear refinement.

TABLE I
INCREMENTAL REGISTRATION RESULTS

Camera	PnP corr.	RANSAC inliers	Final center (C_x, C_y, C_z)
3	329	6	(1.682, 0.228, 1.213)
4	614	535	(1.556, 0.292, 1.117)
5	640	486	(2.215, 0.250, 2.194)

In each case, the non-linear PnP refinement shifts the camera slightly from its linear DLT estimate, as visible in Figs. 6–8. The red (linear) and blue (non-linear) markers are consistently close but distinguishable, with the non-linear center displaced toward lower Z values. This systematic shift is expected: the linear DLT minimises an algebraic cost that does not directly correspond to pixel reprojection error, whereas the LM step explicitly minimises the geometric distance between projected and observed points.

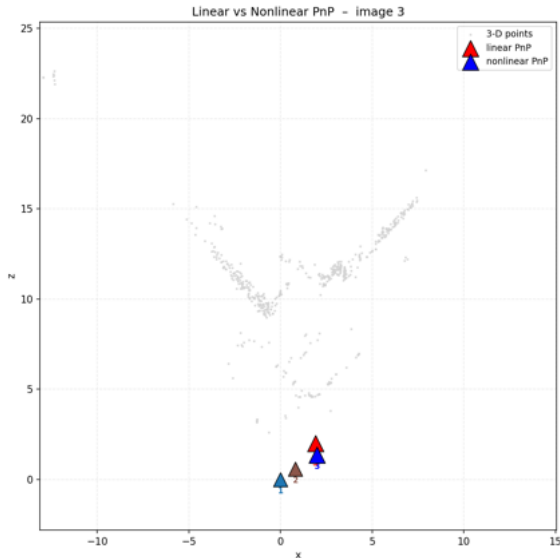


Fig. 6. Linear (red) vs. non-linear (blue) PnP for camera 3. Cameras 1 and 2 (bootstrap) shown in their final tab10 colours.

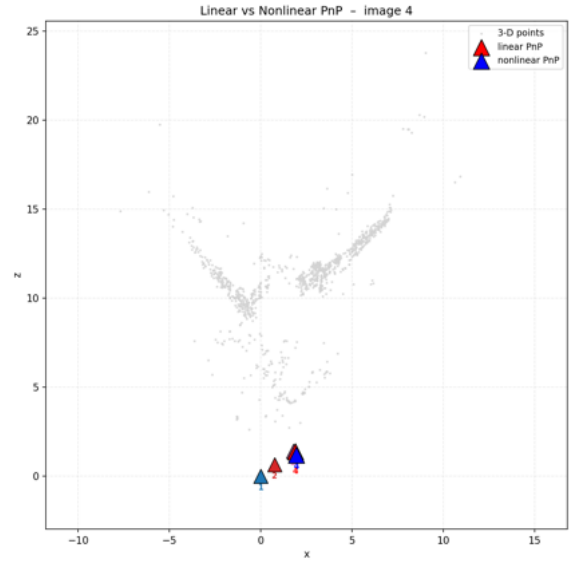


Fig. 7. Linear (red) vs. non-linear (blue) PnP for camera 4. Cameras 1–3 shown in their final tab10 colours.

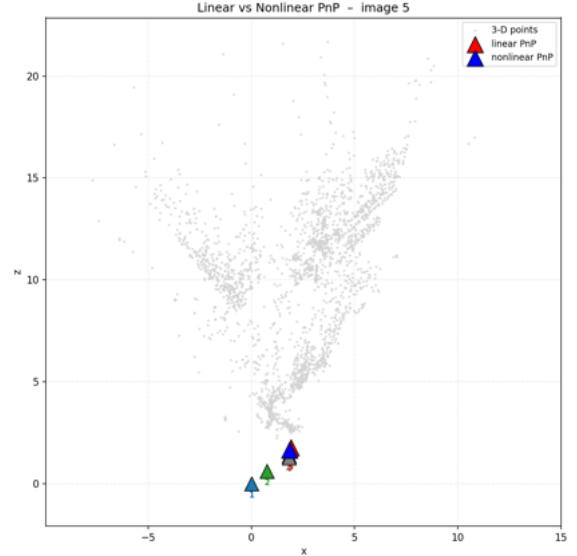


Fig. 8. Linear (red) vs. non-linear (blue) PnP for camera 5. Cameras 1–4 shown in their final tab10 colours.

D. Per-Camera Reprojection

At each registration step, new 3-D points are triangulated between the newly registered camera and each previously registered camera, then refined by non-linear triangulation. The reprojection quality of these new points provides a direct assessment of how well each newly registered camera integrates with the existing reconstruction.

For the cam2–cam3 pair (Fig. 9), the linear and non-linear results are already in reasonable agreement, with the reprojected crosses clustering tightly around their observed counterparts across both the building facade and the surrounding scene. A moderate improvement is visible in the non-linear row, particularly along the vertical window mullions

where depth is harder to recover from a nearly fronto-parallel viewpoint.

The cam3–cam4 pair (Fig. 10) reveals a more pronounced difference between the two estimates. The linear reprojection (top row) shows a visible scatter of red crosses displaced from their green observations, especially in the lower half of both images. The non-linear refinement (bottom row) substantially corrects this, bringing the projected points into close alignment with the observations. This improvement is attributable to the larger depth variation in the cam3–cam4 baseline, which makes the linear error of the DLT more significant.

By the cam4–cam5 pair (Fig. 11), the pipeline has accumulated a dense and accurate 3-D cloud, and both the linear and non-linear results show tight reprojection quality. The non-linear row achieves near-perfect alignment across the entire facade, with the improvement over the linear estimate concentrated at the image borders where lens distortion and out-of-plane scene geometry are more challenging.



Fig. 9. Reprojection overlay for the cam2–cam3 pair after camera3 registration. Linear (top) vs. non-linear (bottom).

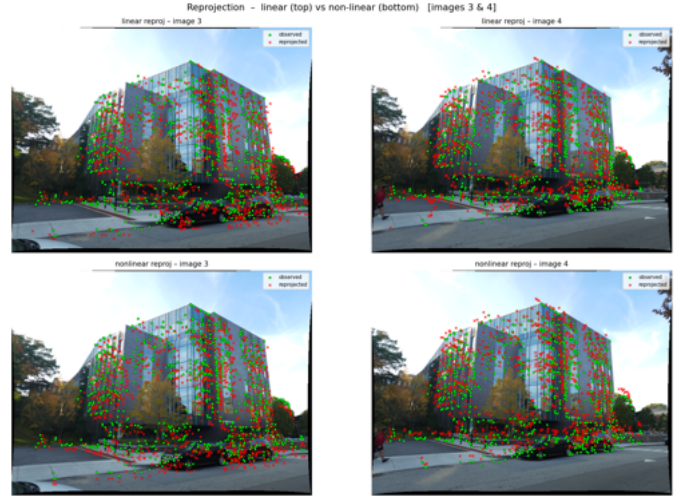


Fig. 10. Reprojection overlay for the cam3–cam4 pair after camera4 registration. The larger error spread in the linear top row is substantially corrected by non-linear refinement (bottom).

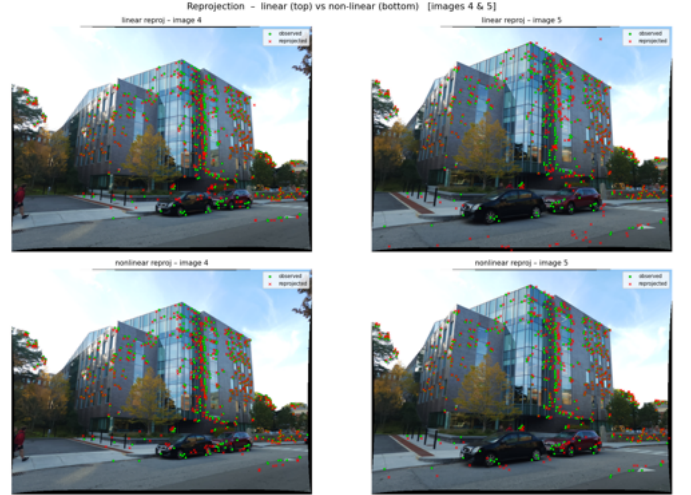


Fig. 11. Reprojection overlay for the cam4–cam5 pair after camera5 registration. Non-linear refinement (bottom) achieves tight green-red alignment across the full facade.

E. Bundle Adjustment Convergence

Three rounds of bundle adjustment are performed, one after each new camera is registered. As shown in Table II, both the point count and observation count grow substantially with each round, reflecting the cumulative triangulation of new tracks. The raw final cost rises from 546 to 5,615, but this increase is driven entirely by the growing number of residual terms: the cost per observation remains low and consistent across all three rounds, confirming that the reconstruction quality does not deteriorate as the reconstruction expands.

The geometric effect of BA is most clearly visible in Fig. 12, which overlays the pre-BA (blue) and post-BA (red) point clouds. Before optimisation, the cloud contains noticeable outlier streaks extending away from the main building surface, particularly along the upper portion of the facade where depth

is poorly constrained by the bootstrap pair alone. After BA, these streaks collapse into a tighter surface, and the two-wing V-shape of the building becomes sharper and more well-defined. The camera positions (triangle markers) remain stable across BA rounds, as a direct consequence of the gauge-fixing strategy that anchors camera 0 at the origin and prevents the reconstruction from drifting under the global similarity ambiguity.

TABLE II
BUNDLE ADJUSTMENT CONVERGENCE PER ROUND

After cam.	Cameras	Points	Obs.	Final cost
3	3	777	1,644	546
4	4	1347	3,306	1,968
5	5	1674	4,474	5,615

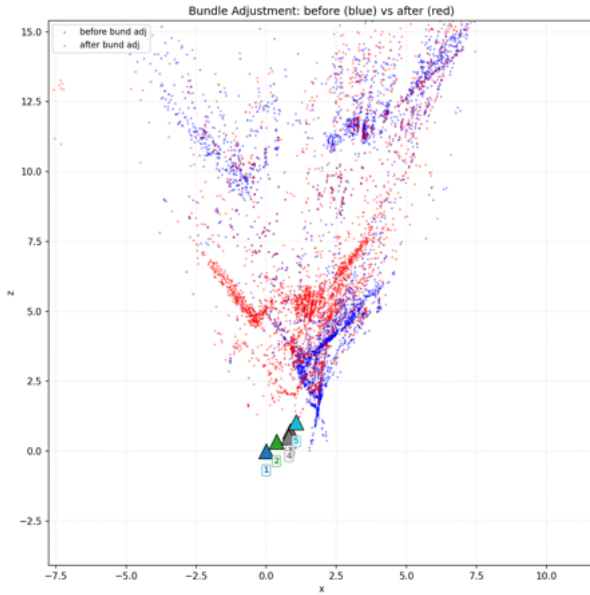


Fig. 12. Before-BA (blue) vs. after-BA (red) point clouds (X-Z top-down view) with all 5 camera positions. BA consolidates scattered blue points onto a tighter surface consistent with the building geometry, particularly along the upper portions of both wings.

F. Final Reconstruction

The completed reconstruction comprises 3,061 3-D points and five registered cameras, as shown in Fig. 13. In the top-down X-Z view, the recovered point cloud traces a distinct V-shape whose two arms correspond precisely to the two glass-and-concrete wings of Unity Hall meeting at a central corner. This structure is geometrically consistent with the known architecture of the building and confirms that the pipeline has correctly recovered the scene layout rather than fitting an arbitrary configuration to the image data.

The five camera centers are arranged in a compact arc at the base of the V, spanning a horizontal baseline of $\Delta C_x \approx 2.2$ and a depth range of $\Delta C_z \approx 2.2$. The optical-axis arrows point outward and upward from the building, correctly reflecting the

direction in which each camera was facing during acquisition. The smooth progression of camera positions from left to right, Cam 1 at the origin through Cam 5 at (2.2, 0.25, 2.2), is consistent with a photographer walking along the building facade and sweeping around the corner, which matches the visual content of the image sequence.

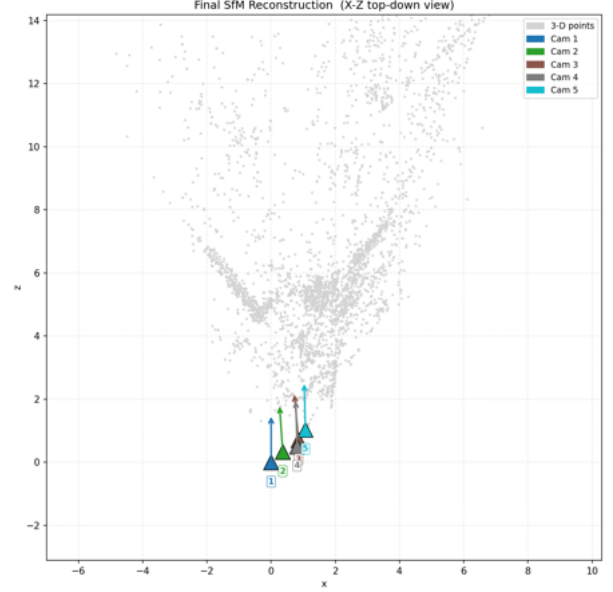


Fig. 13. Final SfM reconstruction (X-Z top-down view). Grey: 3,061 recovered 3-D points. Coloured triangles with arrows: 5 camera positions and optical axes (Cam 1 blue, Cam 2 green, Cam 3 brown, Cam 4 grey, Cam 5 cyan). The V-shape corresponds to the two wings of Unity Hall.

IX. DISCUSSION

a) PnP quality across cameras: Camera 3 obtained only 6 RANSAC inliers from 329 correspondences, yet the non-linear refinement converged to a geometrically reasonable pose (1.682, 0.228, 1.213) because the 6 inliers were well-distributed. Cameras 4 and 5 achieved 535 and 486 inliers respectively after the sign disambiguation fix was applied to LinearPnP, confirming that the cheirality correction was essential for the later cameras where the DLT ambiguity was more pronounced due to the larger translation from the bootstrap frame.

b) Reprojection quality: The reprojection figures reveal that the bootstrap pair (images 1 and 2) achieves the best alignment, with red crosses almost perfectly centered on the green observations (Fig. 5). The cam 3–cam 4 pair (Fig. 10) shows a larger spread in the linear top row, reflecting accumulated noise from the growing 3-D cloud; the non-linear refinement substantially corrects this. The cam 4–cam 5 pair (Fig. 11) shows the tightest overall alignment, likely because these adjacent images share the largest overlap and highest inlier count.

c) Bundle Adjustment behavior: The rising BA cost across rounds (546 $\rightarrow$ 1,968 $\rightarrow$ 5,615) reflects the growing number of observations rather than deteriorating quality, as the cost per observation decreases as more of the scene is

explained by a consistent 3-D model. The before/after comparison in Fig. 12 shows that BA successfully consolidates the scattered pre-BA cloud (blue) into a more coherent structure (red), particularly visible in the tightening of the two building wings.

X. CONCLUSION

This project implemented a complete classical incremental SfM pipeline from scratch and validated it on a five-image outdoor sequence of Unity Hall at WPI. Starting from pre-computed SIFT descriptors, the pipeline successfully processed all the image pairs through RANSAC filtering, estimated a geometrically correct fundamental matrix whose epipolar lines converge at $\mathbf{e}_1 \approx (1155, 440)$ and $\mathbf{e}_2 \approx (1038, 424)$, and bootstrapped a reliable 3-D seed from the first image pair. Incremental registration of cameras 3–5 via PnP-RANSAC and non-linear refinement grew the sparse model from 556 bootstrap points to a final cloud of 3,061 points, with three rounds of gauge-fixed bundle adjustment progressively tightening the reconstruction to final costs of 546, 1,968, and 5,615 respectively. The recovered geometry reproduces the V-shaped two-wing facade of the building, and the five camera centers trace a smooth arc consistent with the acquisition trajectory, together confirming that the pipeline has correctly solved the underlying reconstruction problem.

A central lesson of this work is that the gap between a theoretically correct SfM formulation and a reliably working implementation lies in a small number of numerical and geometric details. Cheirality-based sign disambiguation in the DLT-PnP solver prevented cameras from being mirrored behind the scene, which would otherwise have caused bundle adjustment to collapse all poses onto a degenerate point. Freezing camera 0 throughout BA removed the 7-DoF gauge ambiguity that allowed the optimiser to shift the entire reconstruction without improving geometry. Together these fixes illustrate that in geometric computer vision, theoretical correctness is necessary but not sufficient: careful attention to units, degeneracy conditions, and initialisation quality is what makes the difference between a system that appears to work and one that can be trusted.

The relevance of this pipeline extends well beyond the five images studied here. Classical incremental SfM is the algorithmic foundation of large-scale systems such as Building Rome in a Day [6], which applied the same core stages to reconstruct entire cities from millions of unstructured internet photographs. The resulting sparse point clouds and calibrated camera poses are not merely intermediate outputs; they provide the essential initialization for dense multi-view stereo, surface reconstruction, and modern neural rendering approaches. For this reason, a clear understanding of incremental SfM remains fundamental for researchers and practitioners operating at the forefront of 3D computer vision.

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